TEST PLAN – E-Commerce Web Application (Amazon-like)

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1. Objective

Validate the end-to-end functionality of an e-commerce platform covering:

- User authentication
- Product discovery
- Cart management
- Checkout and payment
- Order creation and tracking

Ensure the system is **functionally correct, reliable, and handles edge cases without failure**.

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2. Scope

In Scope

- Login & Registration
- Search & Product Catalogue
- Product Details Page
- Cart Management
- Checkout Flow
- Payment Processing
- Order Confirmation & Tracking

Out of Scope

- Seller/Admin functionalities
- Warehouse/logistics backend
- Advanced recommendation algorithms

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3. Test Strategy

3.1 Testing Types

Type	Coverage
Functional Testing	All user flows and validations
Regression Testing	Core flows after changes

Type	Coverage
API Testing	Login, Product, Cart, Payment APIs
UI Testing	Layout, responsiveness, navigation
Performance Testing	Load on search & checkout
Negative Testing	Invalid inputs, failures
Exploratory Testing	Edge cases and real user behaviour

3.2 Test Approach

- Derive test scenarios from:
  - BRD-lite
  - User flow diagram (Login → Payment)
- Cover:
  - Happy paths
  - Edge cases
  - Failure scenarios
- Prioritize high-risk modules:
  - Payment
  - Checkout
  - Cart calculations

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4. Test Scenarios Coverage

Login

- Valid login / invalid login
- Duplicate registration
- Session timeout

Search

- Relevant results for keyword
- No results scenario
- Navigation to product

Product

- Correct product details
- Price consistency
- Add to cart availability

Cart

- Add/remove/update quantity
- Cart persistence after refresh
- Price calculation accuracy
- Out-of-stock handling

Checkout

- Address validation
- Coupon application
- Session expiry handling

Payment (Critical Module)

- Successful payment creates order
- Failed payment does NOT create order
- Retry payment flow
- No duplicate orders

Order Management

- Order confirmation page
- Order ID generation
- Order history visibility
- Status updates (Placed → Shipped → Delivered)

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5. Test Environment

Component	Details
Browser	Chrome
OS	Windows
Tools	Selenium, Postman, JMeter
Test Data	Dummy users, mock products

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6. Test Data

- Valid and invalid user credentials
- Product datasets (available/out of stock)
- Coupon codes (valid/invalid/expired)
- Payment scenarios (success/failure)

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7. Entry & Exit Criteria

Entry Criteria

- Requirements defined (BRD-lite)
- Test cases prepared
- Test environment ready

Exit Criteria

- All critical test cases executed
- No open high-severity defects
- Payment and checkout flows stable

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8. Risk Analysis

Risk	Impact	Mitigation
Payment failure	Revenue loss	Extensive negative testing
Incorrect pricing	Financial defect	Validate calculations
Cart mismatch	User drop	Data consistency checks
Duplicate order	Critical bug	Idempotency testing

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9. Defect Management

- Tool: JIRA / Excel
- Severity Levels:
  - Critical → Payment/Checkout failure
  - High → Cart/Login issues
  - Medium → UI/Minor bugs

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10. Deliverables

- Test Plan
- Test Scenarios
- Test Cases
- Bug Reports
- Automation Scripts
- Test Execution Report

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11. Timeline

Week	Activity
Week 1	Requirement + Test Planning
Week 2	Test Case Creation + Automation
Week 3	Execution + Bug Reporting
Week 4	Regression + Reporting